

Network On Chip



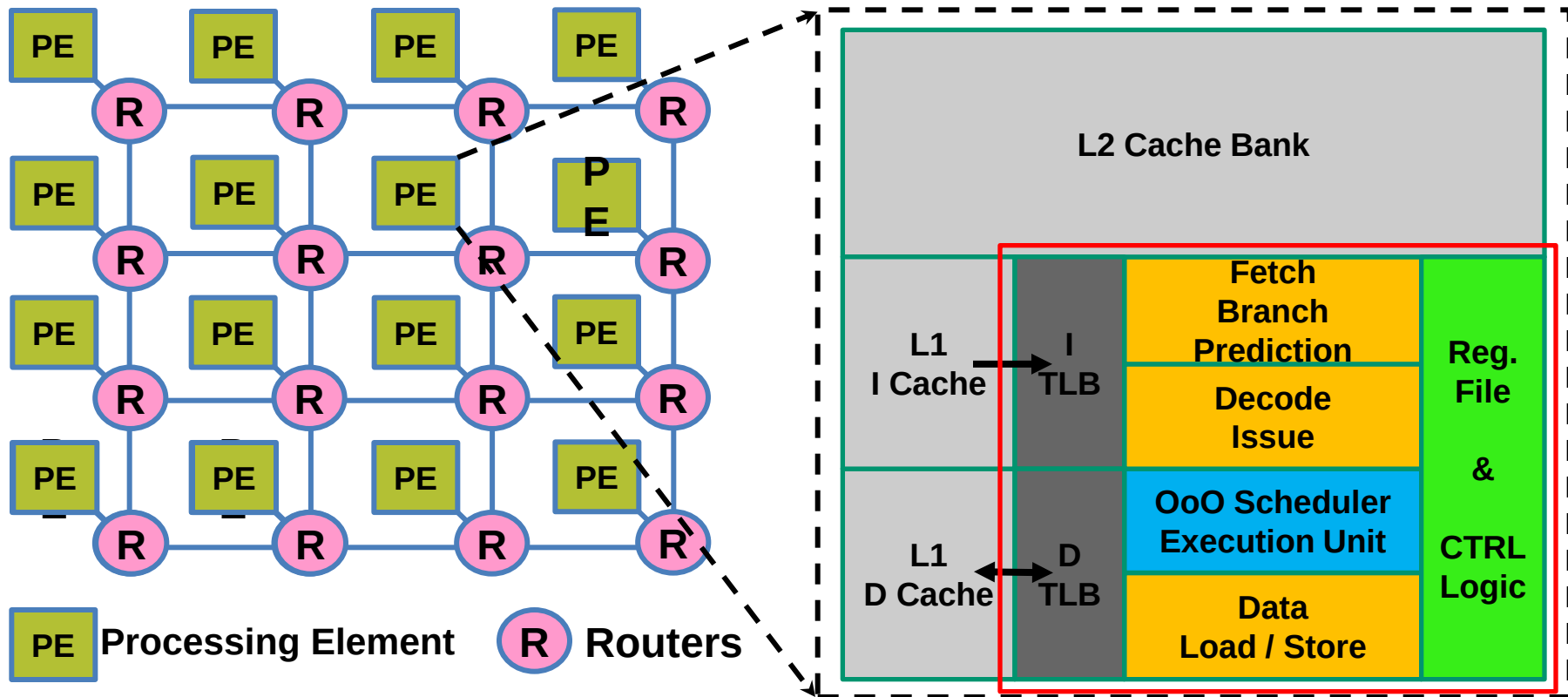
John Jose

Associate Professor

Department of Computer Science & Engineering

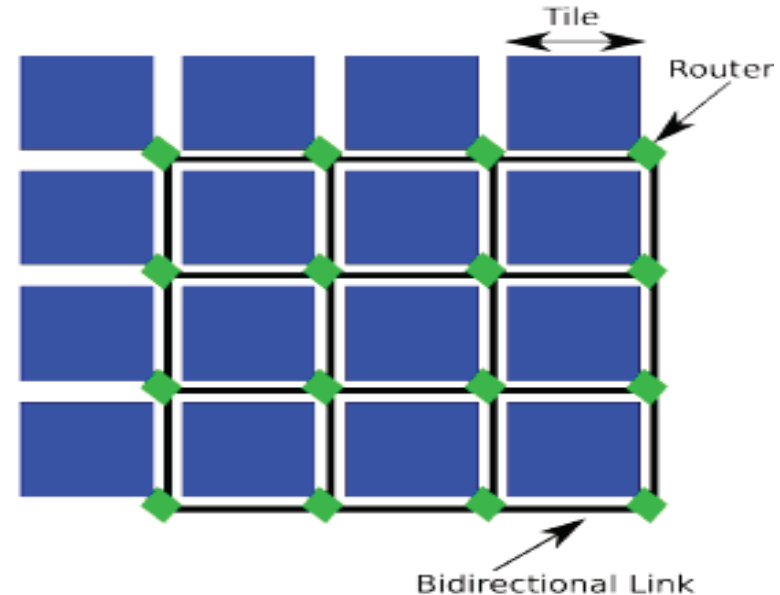
Indian Institute of Technology Guwahati

Tiled Chip Many-Core Processor (TCMP)

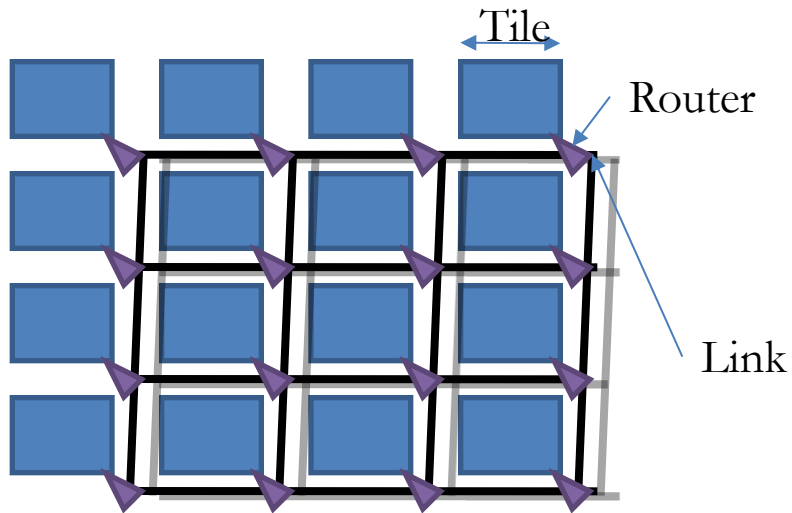


What is NoC ?

- ❖ Processing units interconnected via packet based network
- ❖ Each resource is called as a ***'tile'***
- ❖ All resources organized as rectangular tiles on the chip.
- ❖ Each tile have an address - (X, Y)
- ❖ Tiles interconnected by network of routers
- ❖ Communication by packet transmission



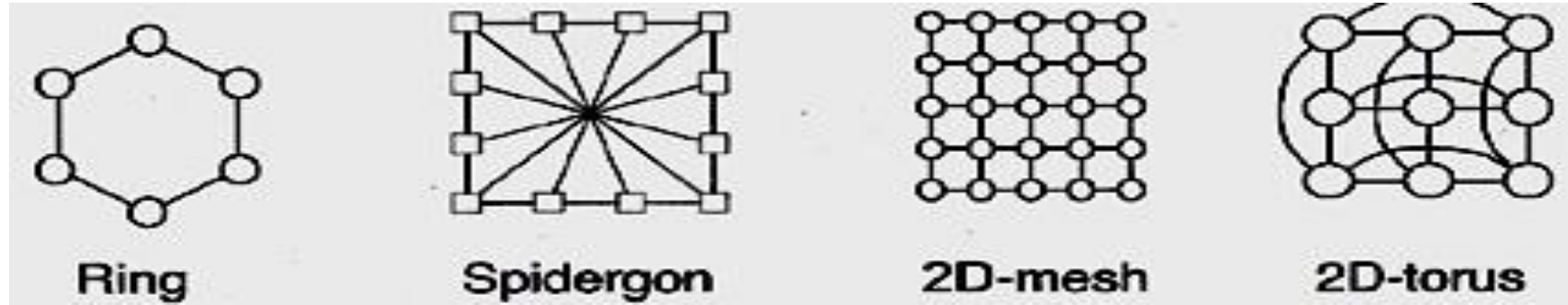
Building Blocks of NoC



- ❖ **Topology**
- ❖ **Routing**
- ❖ **Flow control**
- ❖ **Router micro-architecture**

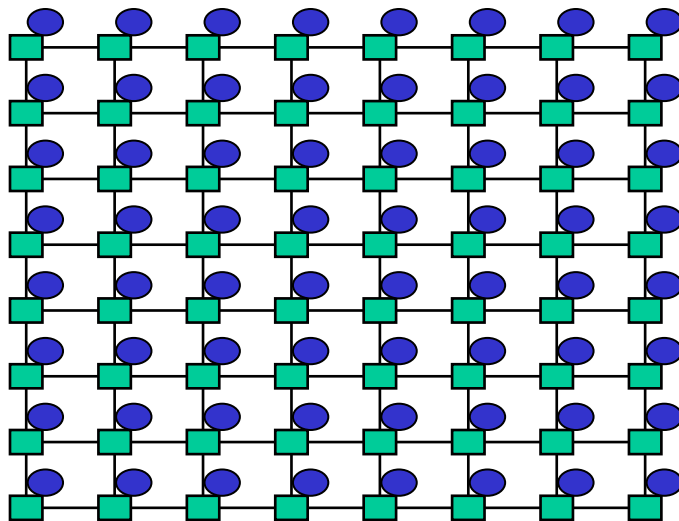
Topology

- ❖ Determines the physical layout and connection pattern between nodes and channels in the network.

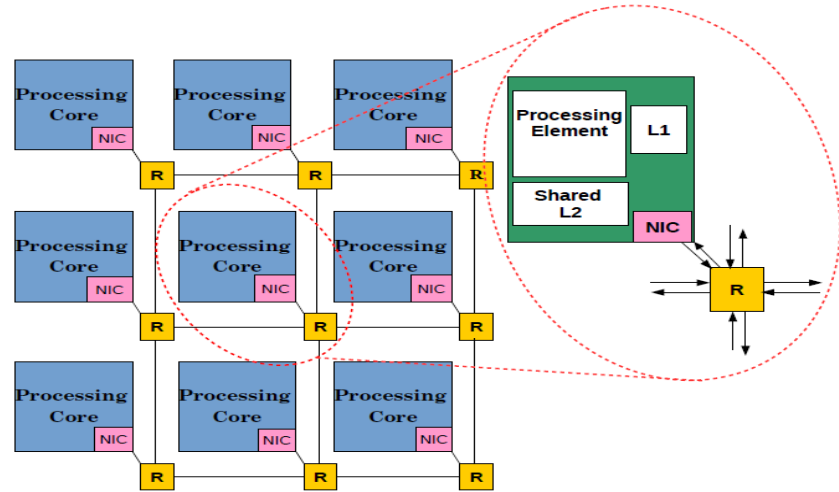
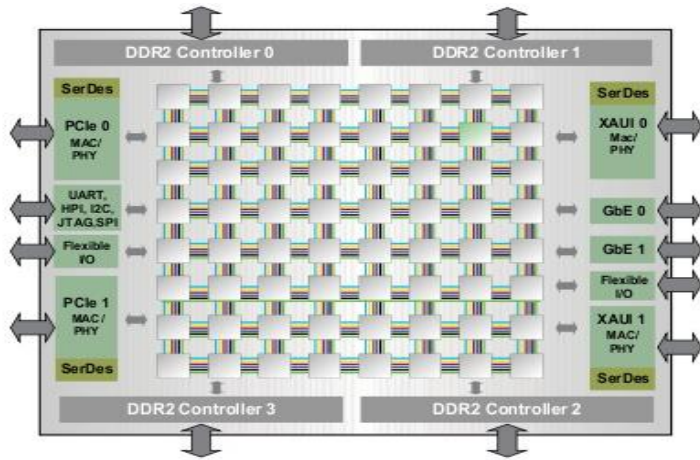


Mesh

- ❖ Each node connected to 4 neighbors (N, E, S, W)
- ❖ Easy to layout on-chip: regular and equal-length links
- ❖ Path diversity: many ways to get from one node to another

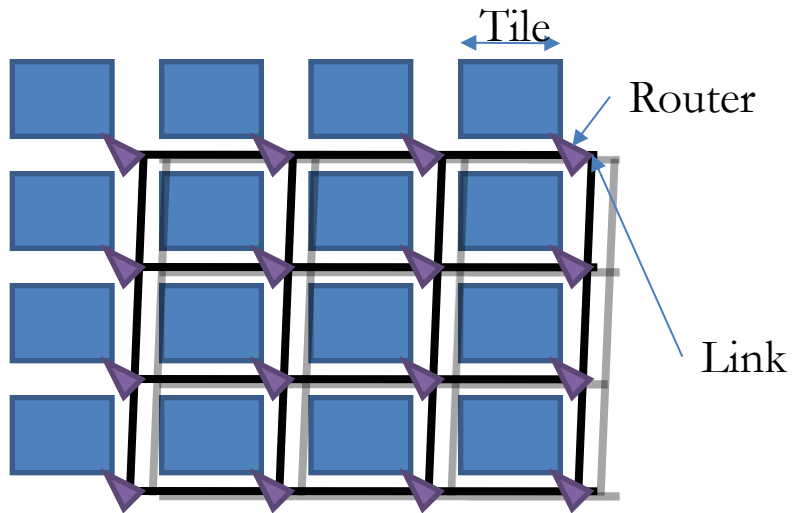


Routers and Tiles



- ❖ East, West, North and South neighbors
- ❖ Packets are divided into flow control units called flits
- ❖ L1 and L2 cache misses create NoC traffic packets

Building Blocks of NoC



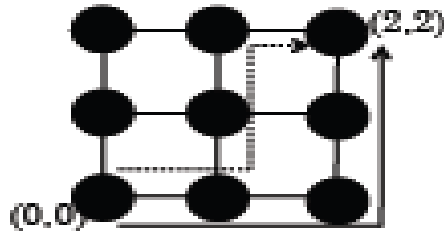
- ❖ **Topology**
- ❖ **Routing**
- ❖ **Flow control**
- ❖ **Router micro-architecture**

Routing Algorithm

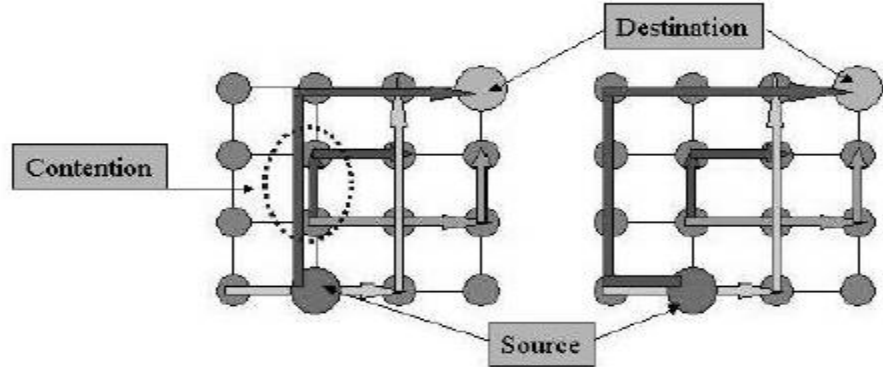
- ❖ Compute the path route for packets to reach destination.
- ❖ Deterministic Routing vs Adaptive Routing
 - ❖ **Deterministic:** always chooses the same path for a communicating source-destination pair
 - ❖ **Oblivious:** chooses different paths, without considering network state
 - ❖ **Adaptive:** can choose different paths, adapting to the state of the network

Minimal & Non-Minimal Routing

- ❖ **Profitable route:** The route that always leads the packet closer to the destination.
- ❖ **Misroute:** A route that leads the packet away from the destination.



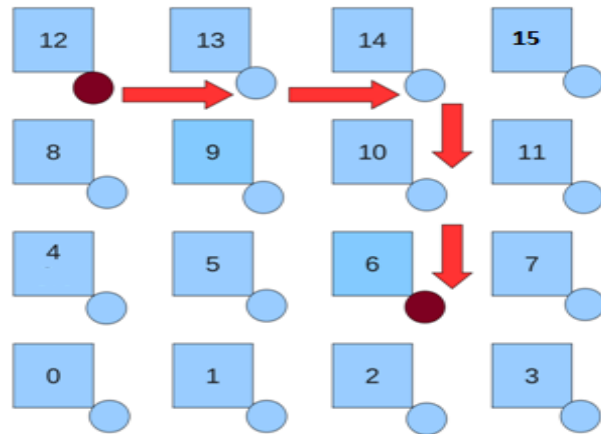
Minimal routing



Non-Minimal routing

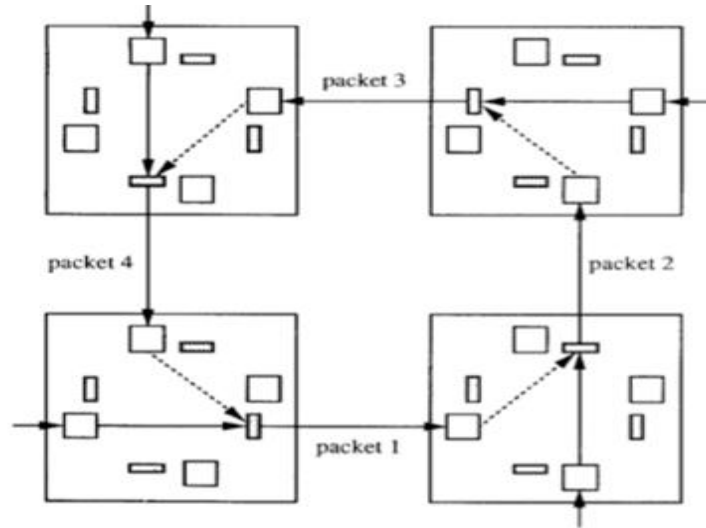
Deterministic Routing

- ❖ All packets between the same (source, dest) pair take the same path
- ❖ Dimension-order routing – Eg. XY routing
 - ❖ First traverse dimension X, then traverse dimension Y
 - ❖ Simple
 - ❖ Deadlock freedom
 - ❖ Could lead to high contention
 - ❖ Does not exploit path diversity



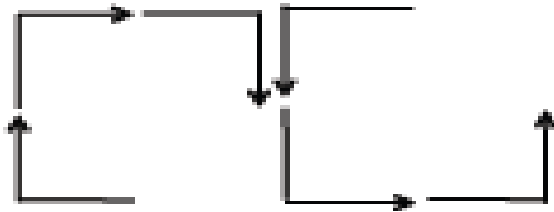
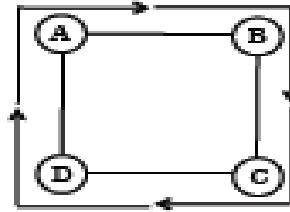
Deadlock

- ❖ No forward progress
- ❖ Caused by circular dependencies on resources
- ❖ Each packet waits for a buffer occupied by another packet downstream

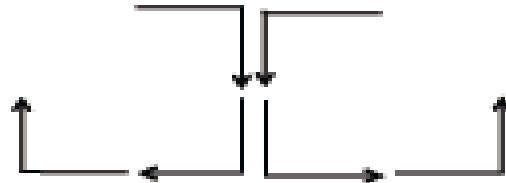


Turn Model to Avoid Deadlock

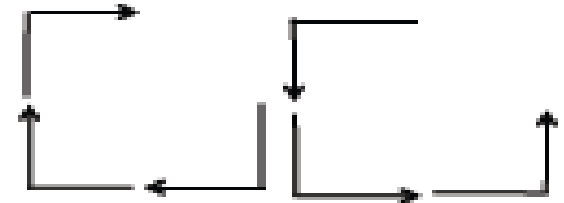
- ❖ Analyze directions in which packets can turn in the network
- ❖ Determine turns the can form cycles
- ❖ Prohibit just enough turns to break possible cycles



West First Turns

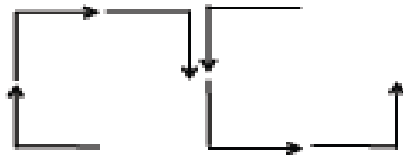


North Last Turns

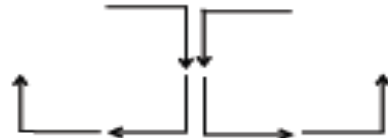


Negative First Turns

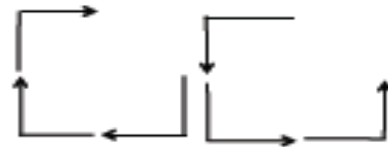
Turn Model to Avoid Deadlock



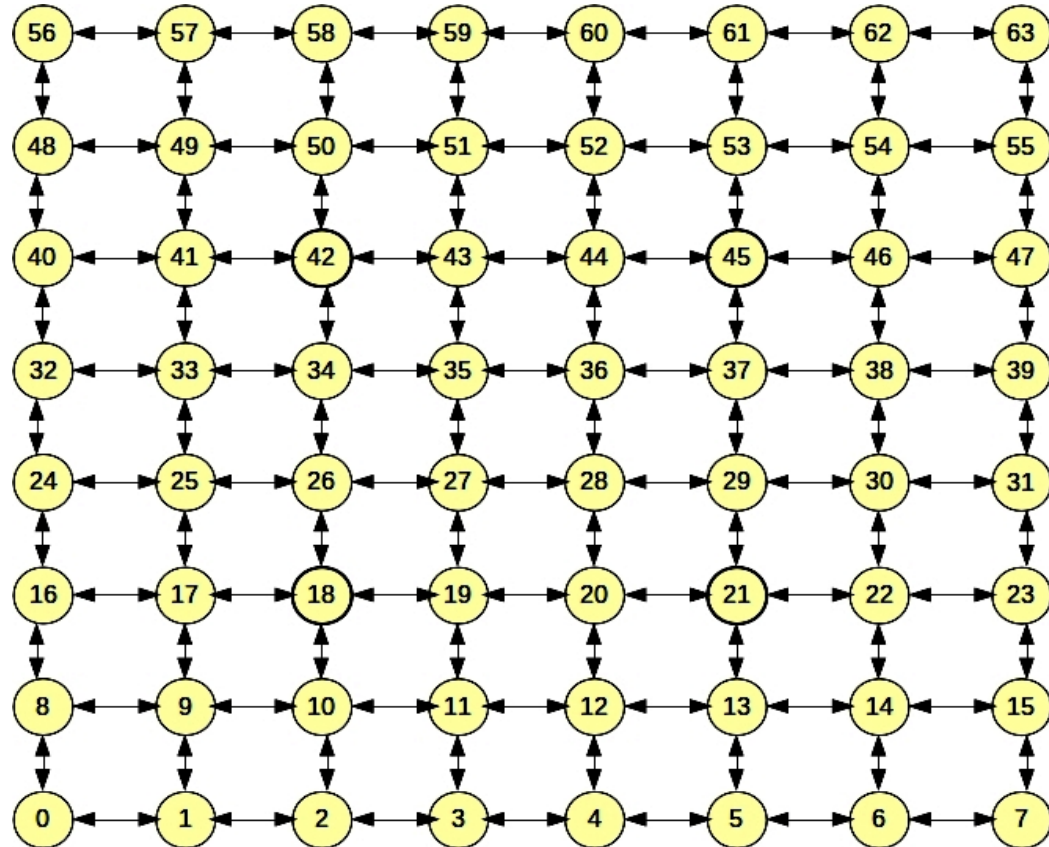
West First Turns



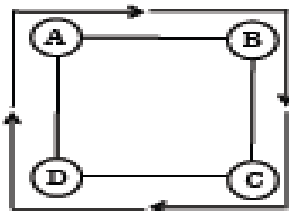
North Last Turns



Negative First Turns

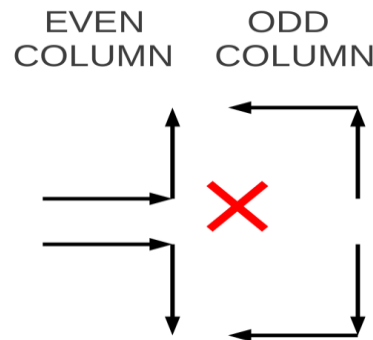
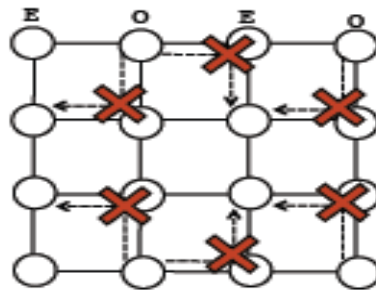
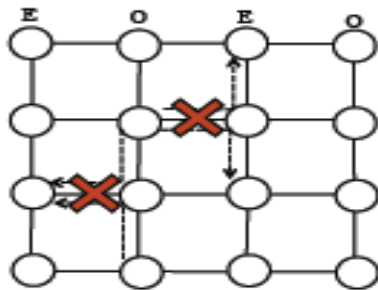


Adaptive Odd-Even Turn Routing

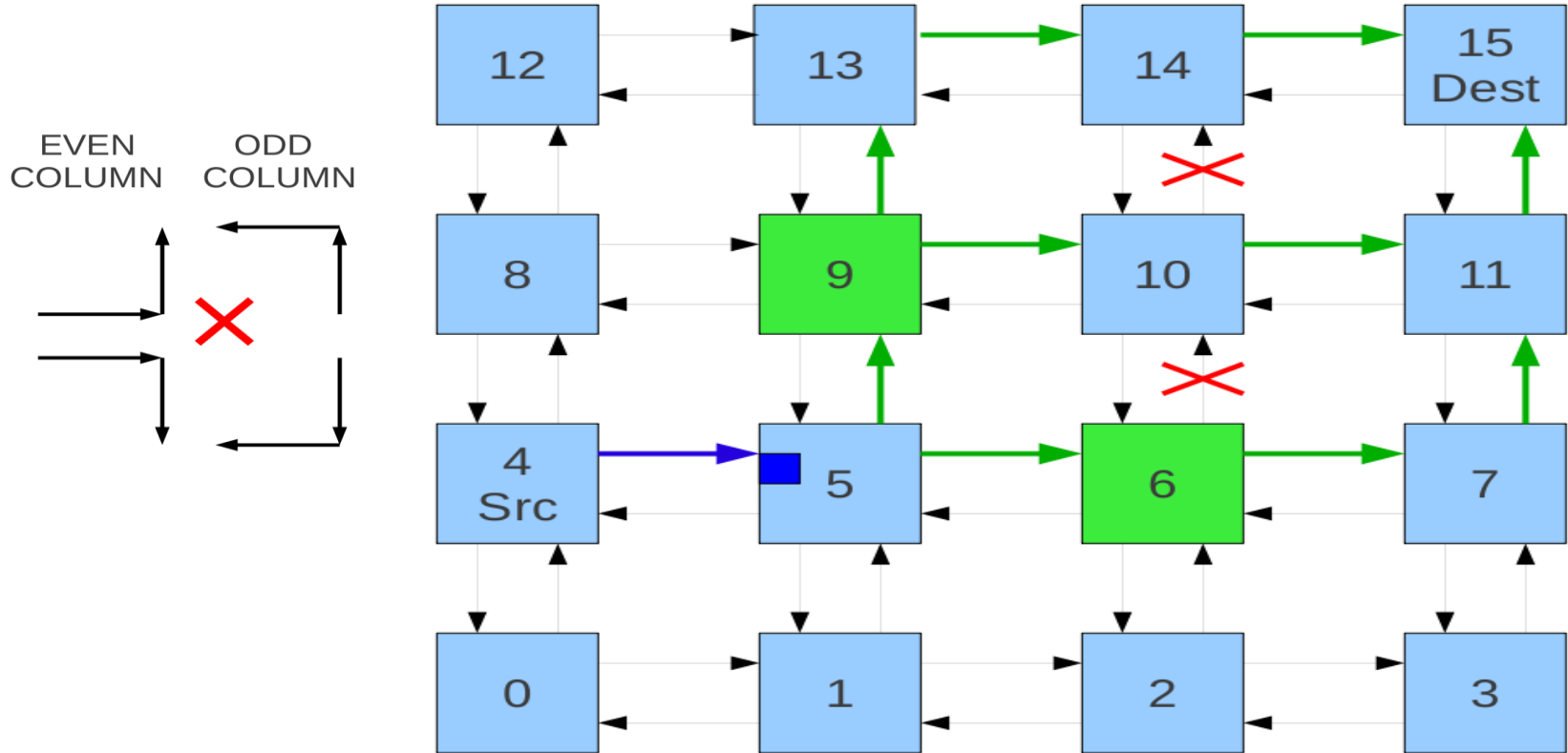


**Non restrictive turns
leads to deadlocks**

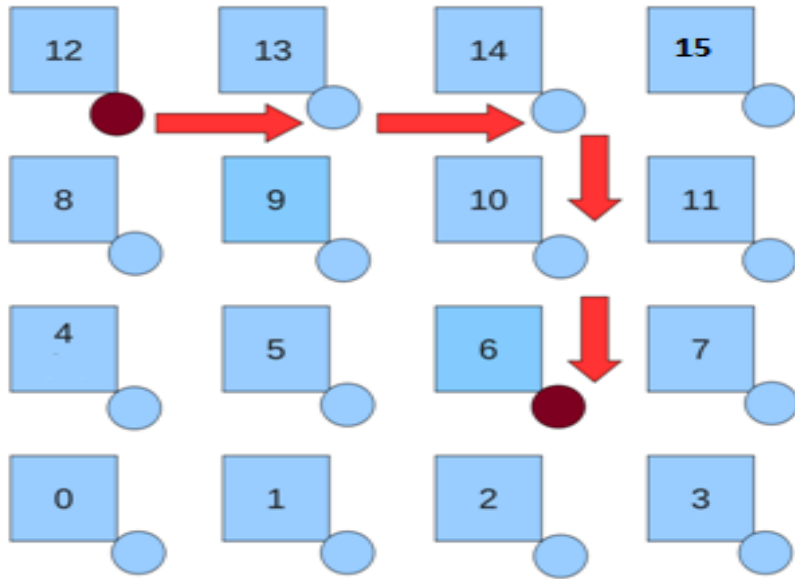
- ❖ Prohibited Turns at certain junctions
- ❖ For nodes in even column EN and ES.
- ❖ For nodes in odd column NW and SW.



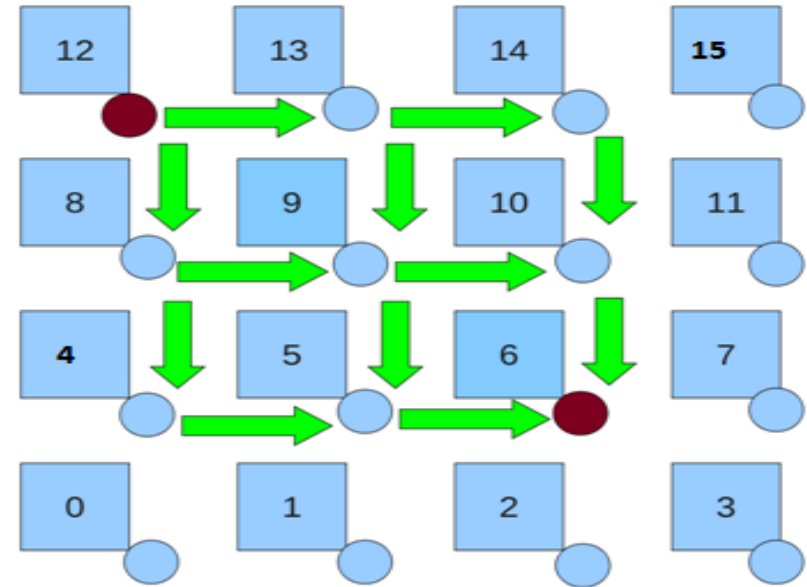
The Minimal Odd-Even Routing



Static vs Adaptive Routing



Static routing – XY routing

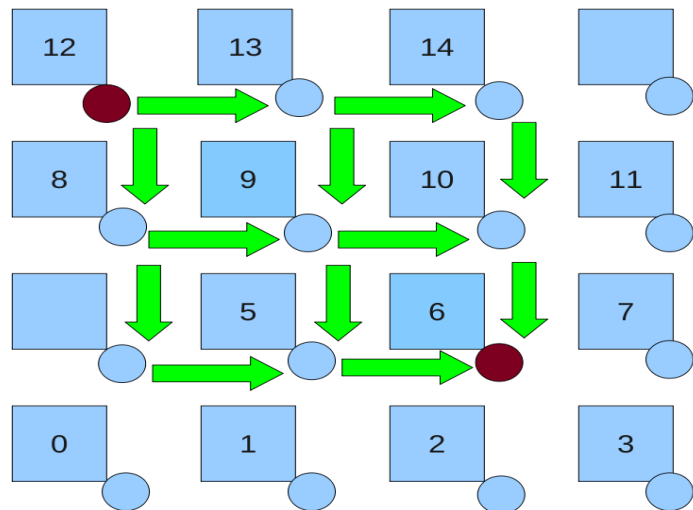


Adaptive Routing

Adaptive Routing

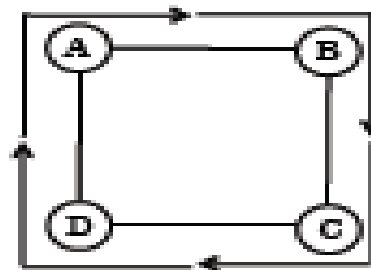
- ❖ Router uses network state (e.g., downstream buffer occupancy) to pick which productive output port to send
- ❖ Productive output port - port that gets the packet closer to its destination
- ❖ Aware of local congestion

Adaptive Routing



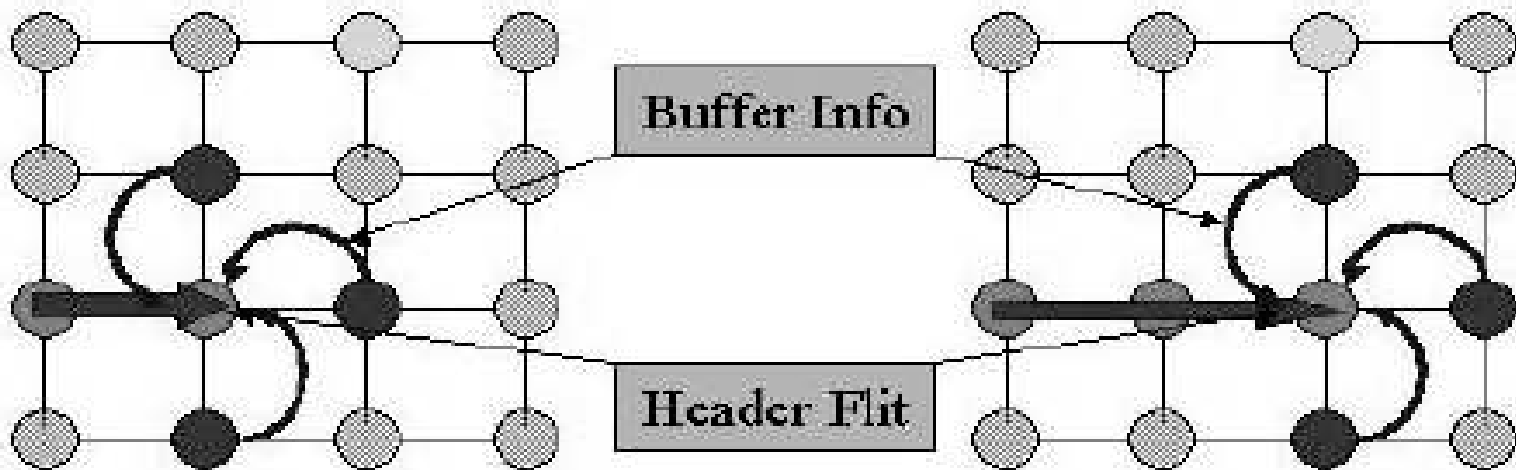
Minimal- Adaptive Routing

- ❖ Local and global information can be exploited for route selection.
- ❖ Preserving inter message order and out of order delivery.
- ❖ Should be deadlock and livelock free.

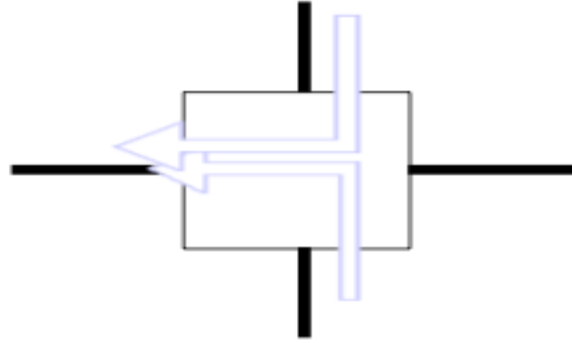


Contention-look-ahead Routing

- ❖ Can foresee the contention of neighbors using control wires.
- ❖ The traffic condition of the neighboring nodes are obtained through the control signal wires.

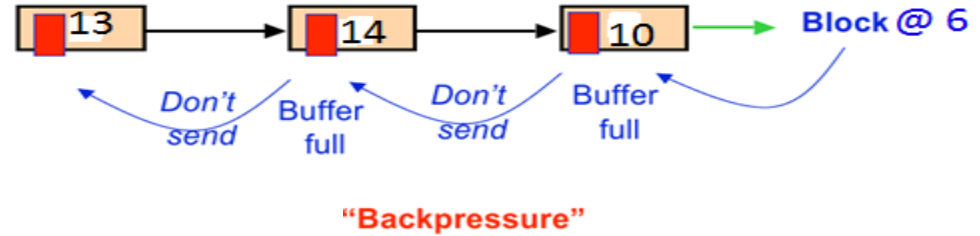
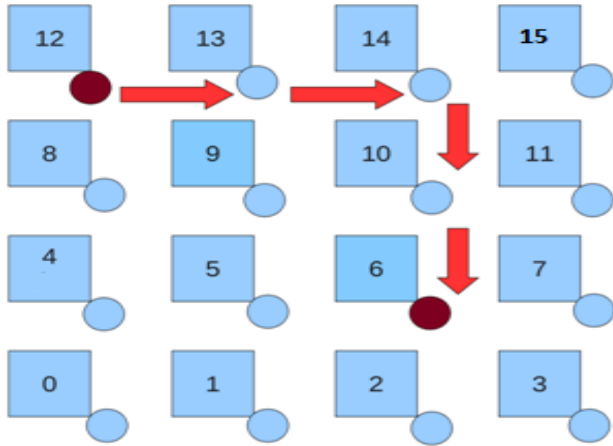


Handling Contention

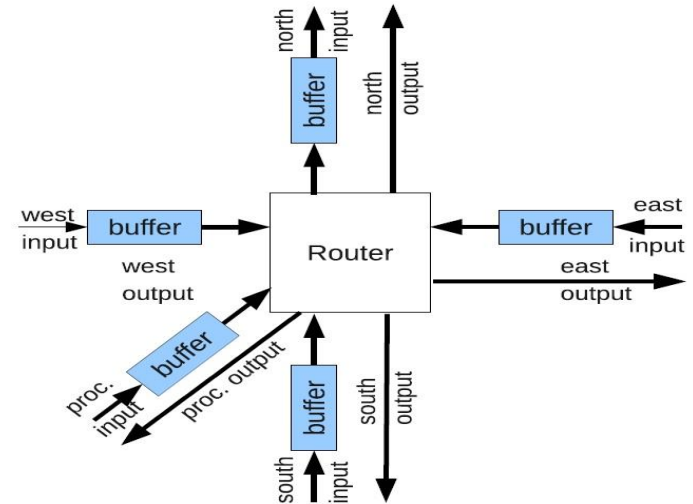


- ❖ When two packets trying to use the same link at the same time ?
 - ❖ Buffer one (widely used mechanism)
 - ❖ Drop one
 - ❖ Misroute one (deflection)

Flow Control

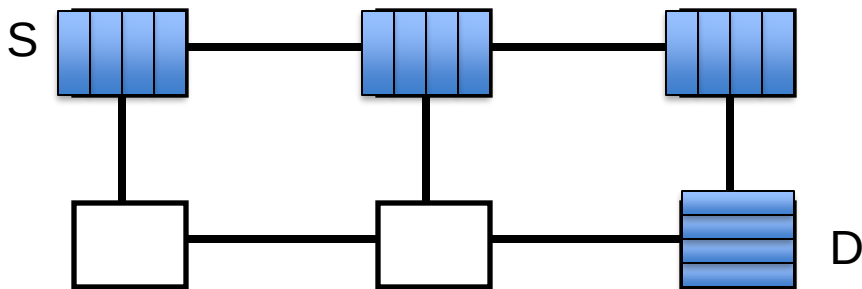


- ❖ Upstream router should know the buffer availability of downstream router.
- ❖ Credit should be exchanged between routers by handshake signals



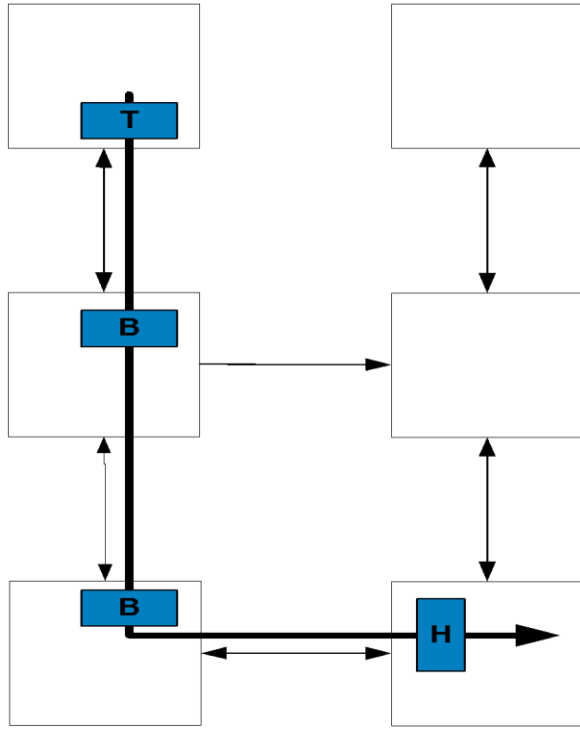
Store and Forward Flow Control

- ❖ Store and Forward Packet based flow control
 - ❖ Packet copied entirely into network router before moving to the next node
 - ❖ Flow control unit is the entire packet
- ❖ Leads to high per-packet latency
- ❖ Requires buffering for entire packet in each node



Can we do better?

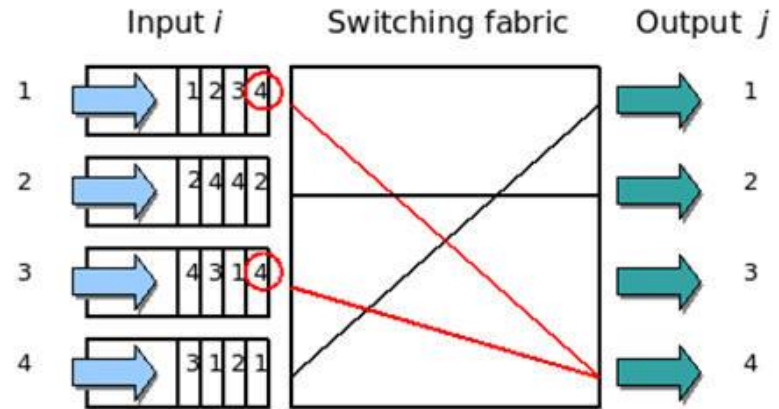
Wormhole Flow Control



- ❖ Packets broken into smaller flits
- ❖ Flits are sent across the fabric in a *wormhole fashion*
 - ❖ Body follows head, tail follows body
 - ❖ Pipelined
 - ❖ If head blocked, rest of packet stops
 - ❖ Routing (src/dest) information only in head
- ❖ Lower latency, efficient buffer utilization
- ❖ Occupies resources across multiple routers

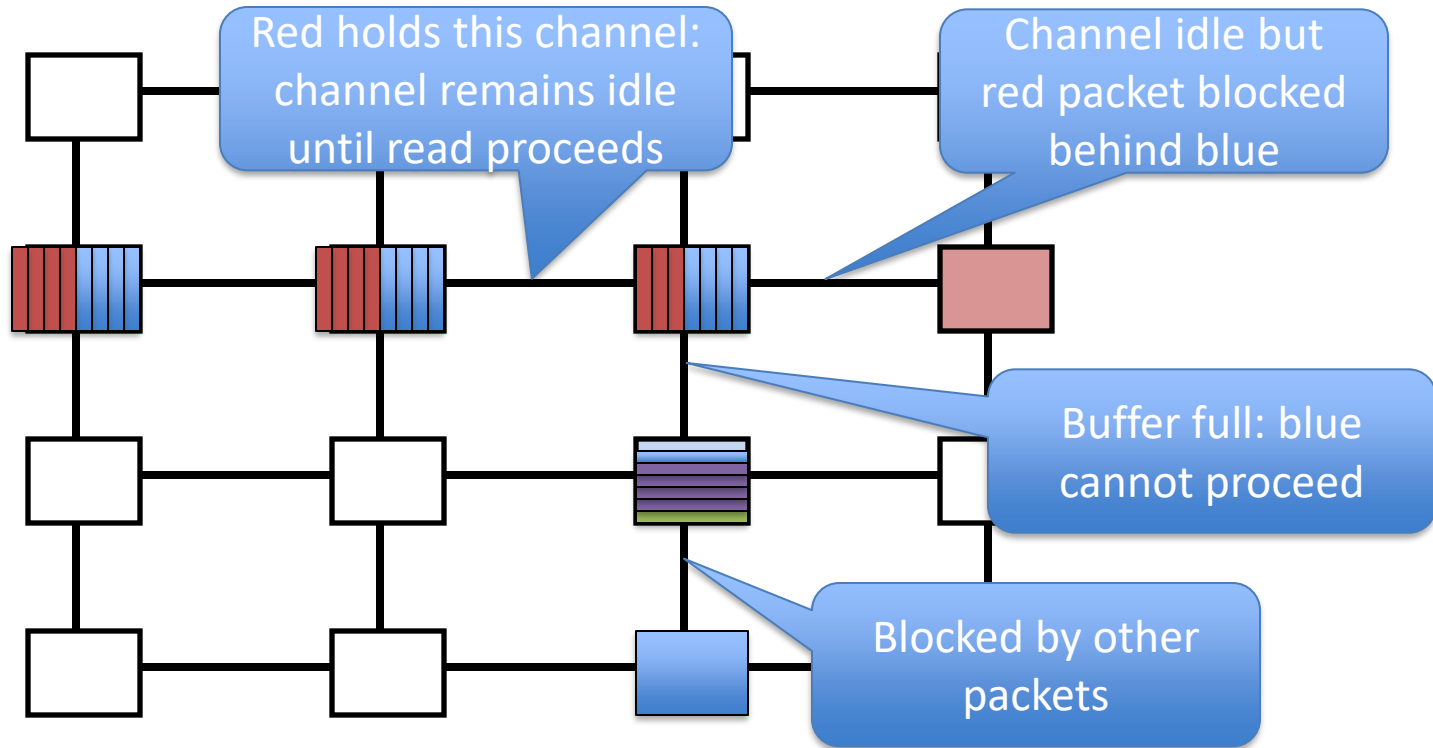
Head of Line Blocking

- ❖ Suffers from **head of line blocking**
 - ❖ If head flit cannot move due to contention, another worm cannot proceed even though links may be idle



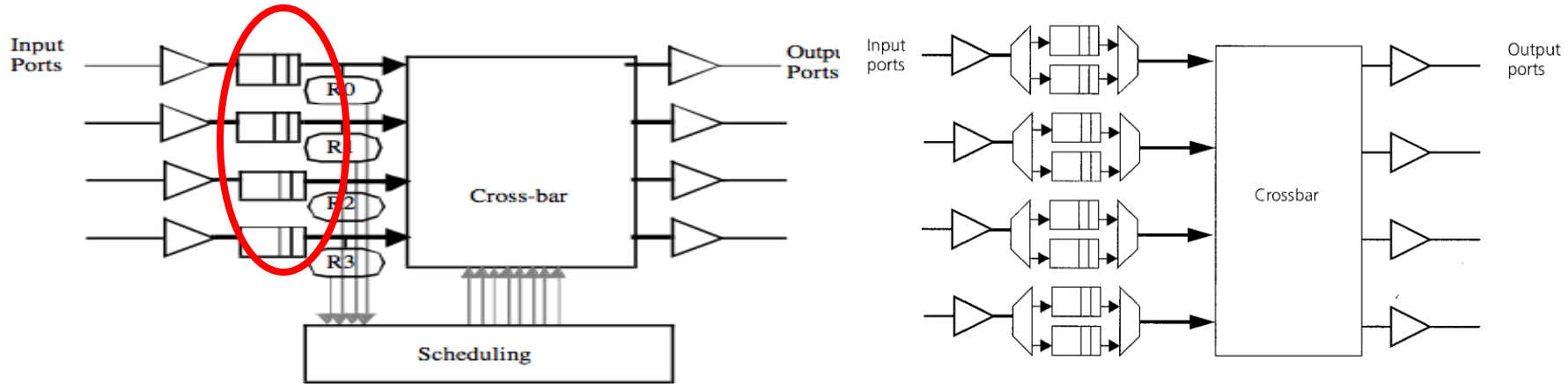
HOL Blocking

Head of Line Blocking

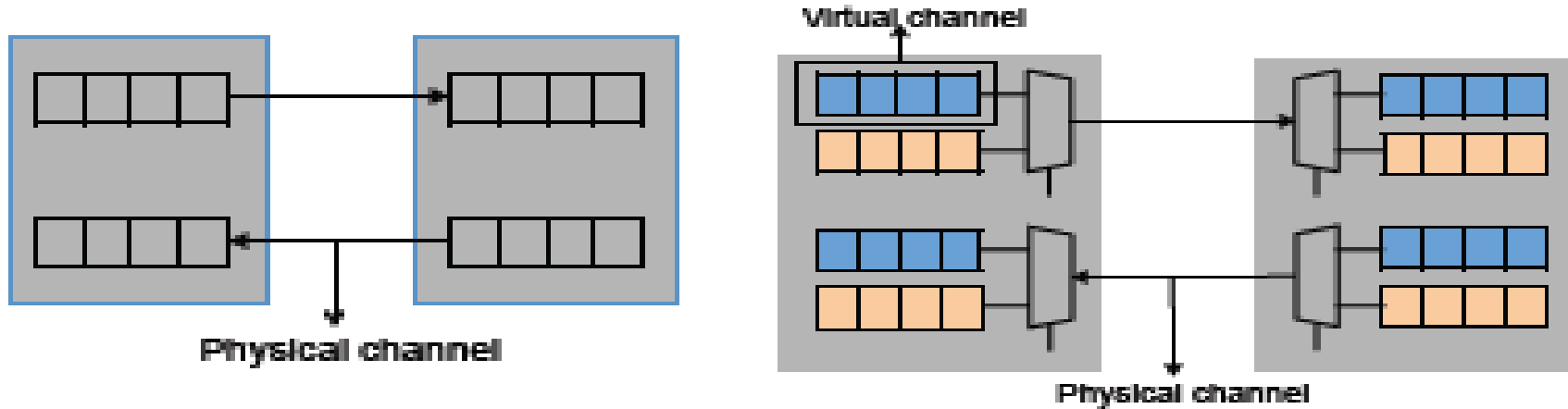


Virtual Channel Flow Control

- ❖ Multiplex multiple channels over one physical channel
- ❖ FIFO buffers replaced with multilane buffers
- ❖ Divide up the input buffer into multiple buffers sharing a single physical channel

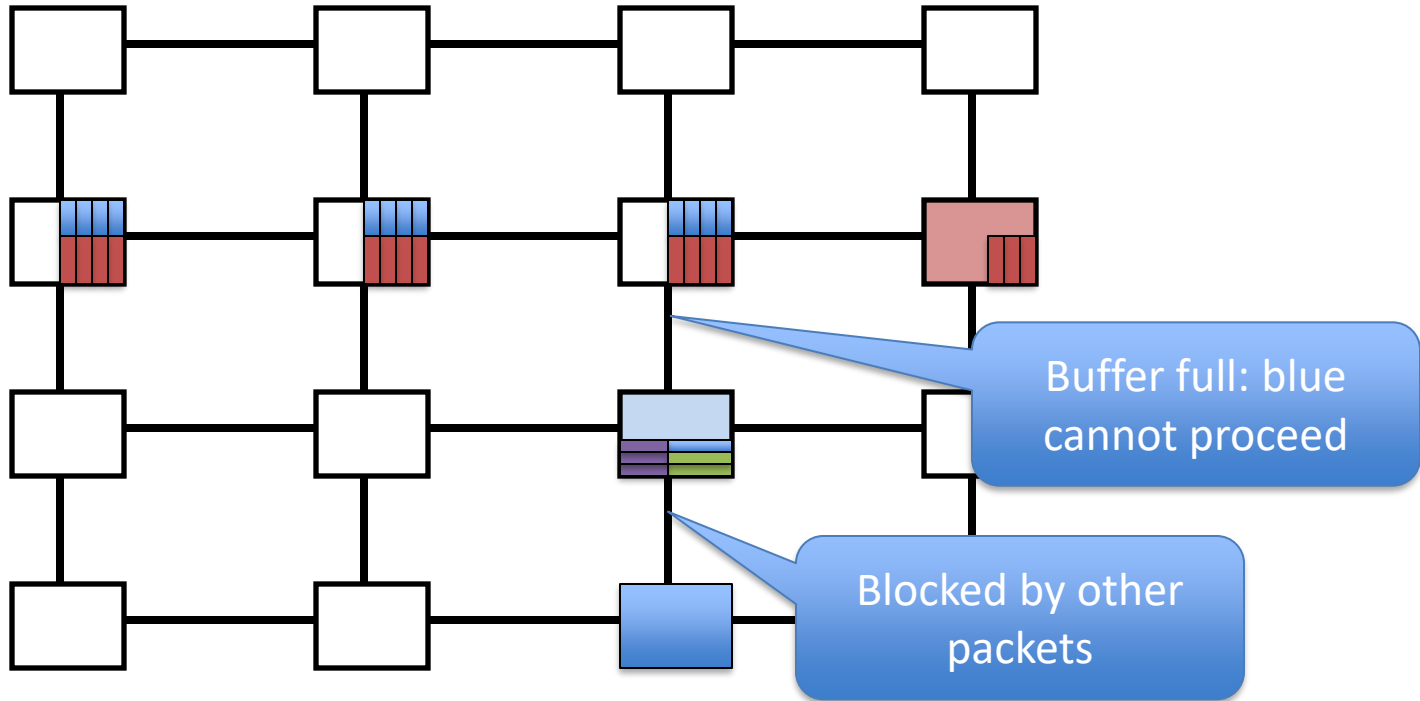


Virtual Channel Flow Control

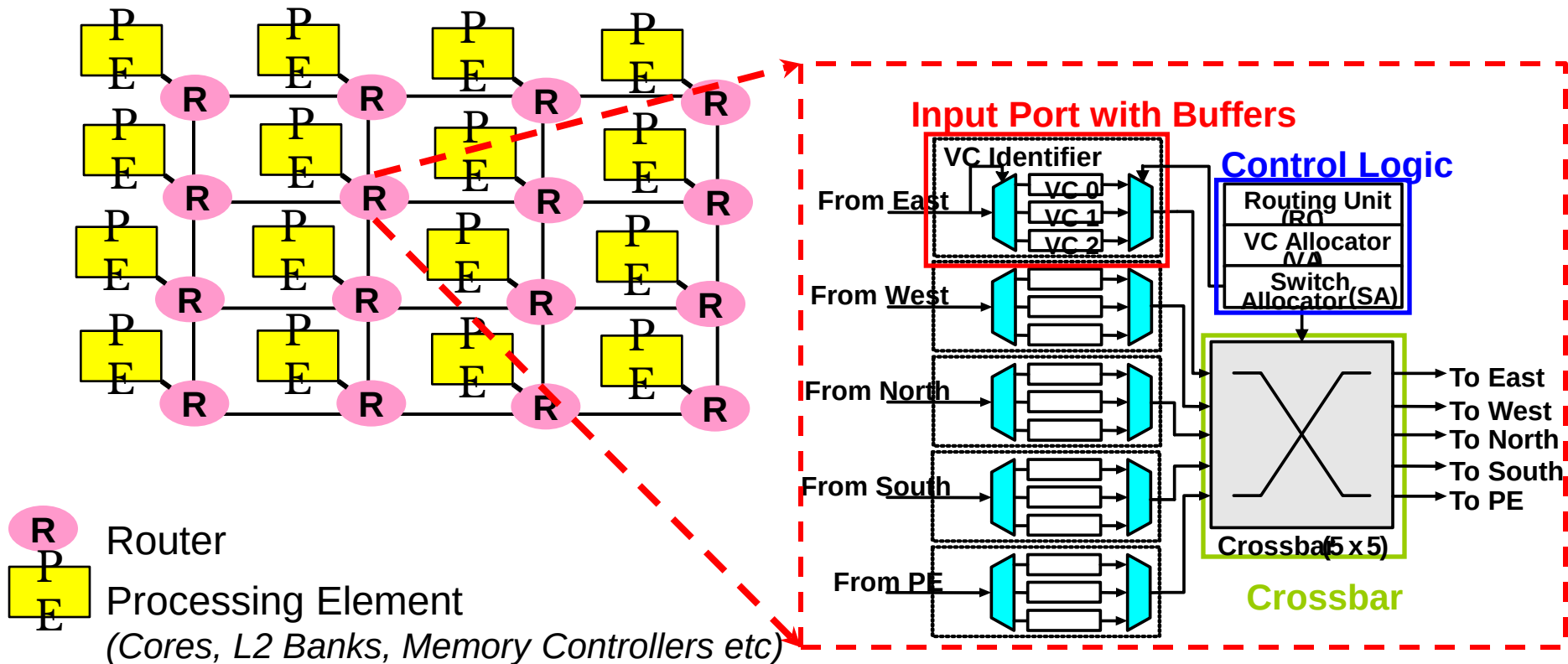


- ❖ VCs are allocated once at each router to the head flit and remaining flits of the packet inherit the same VC
- ❖ Flits of different packets can be interleaved on the same physical channel
- ❖ VCs avoid deadlocks

Virtual Channel Flow Control

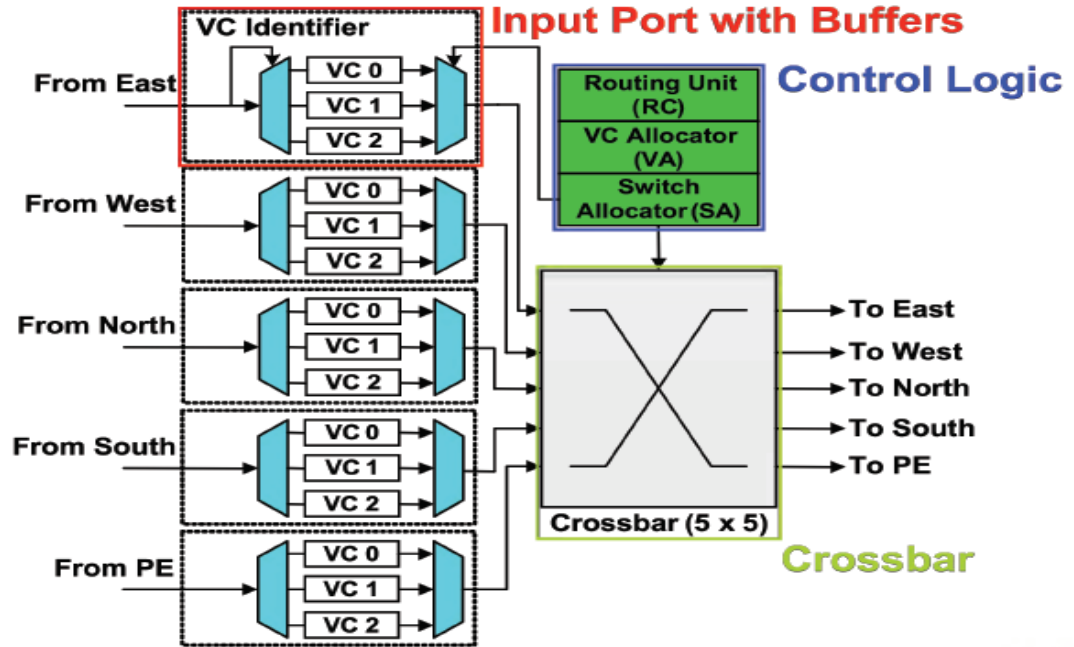


Input Buffered NoC Router

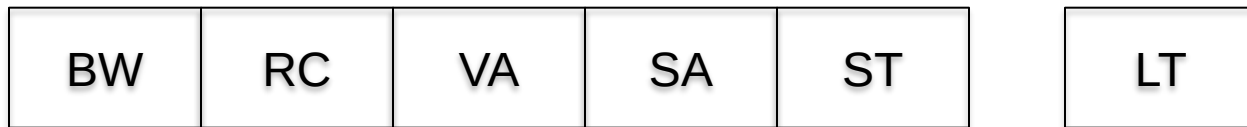


Functions of a Router

- ❖ Buffering of flits
- ❖ Route computation
- ❖ VC allocation
- ❖ Switch Allocation
- ❖ Switch Traversal
- ❖ Link Traversal



Router Pipeline



❖ Five logical stages

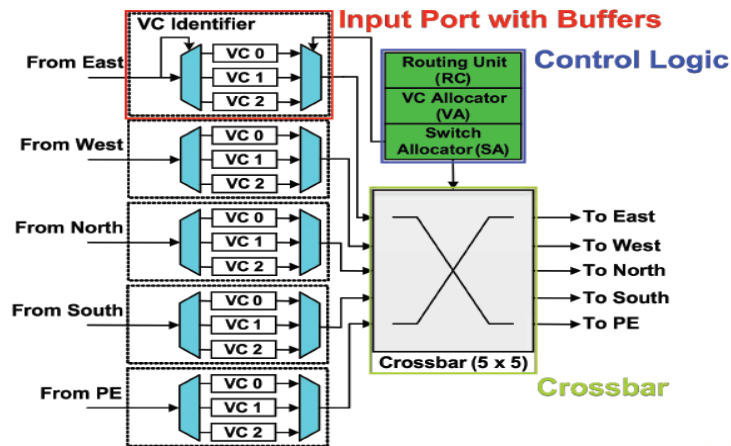
❖ BW: Buffer Write

❖ RC: Route computation

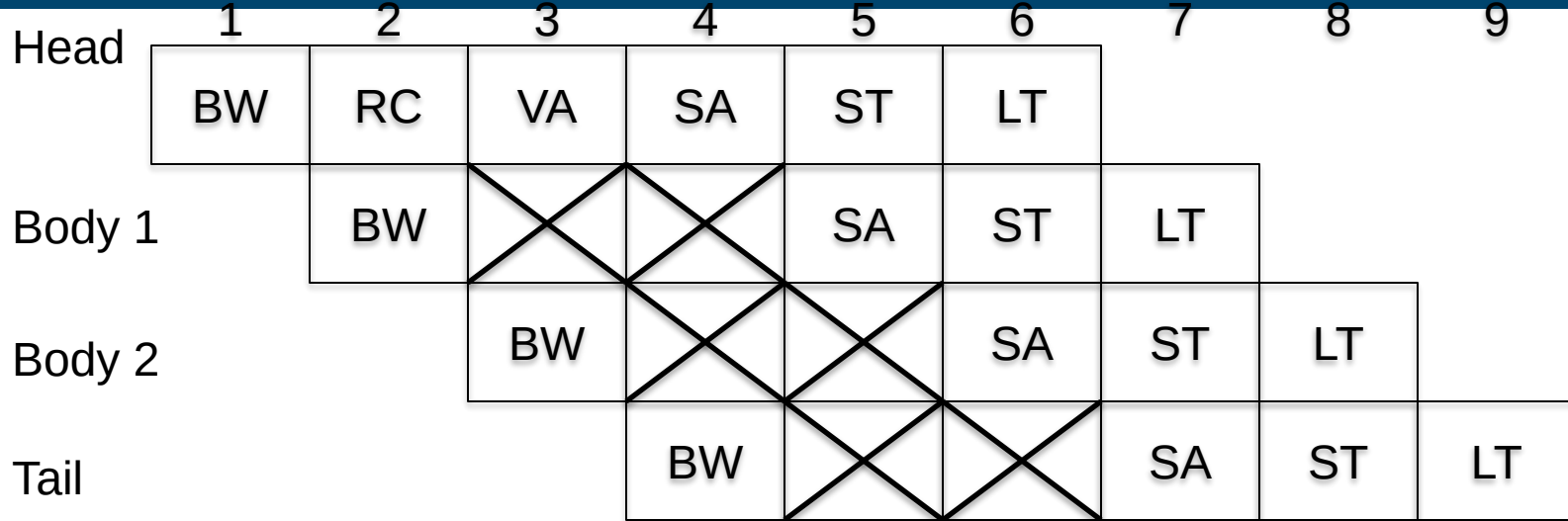
❖ VA: Virtual Channel Allocation

❖ SA: Switch Allocation

❖ ST: Switch Traversal

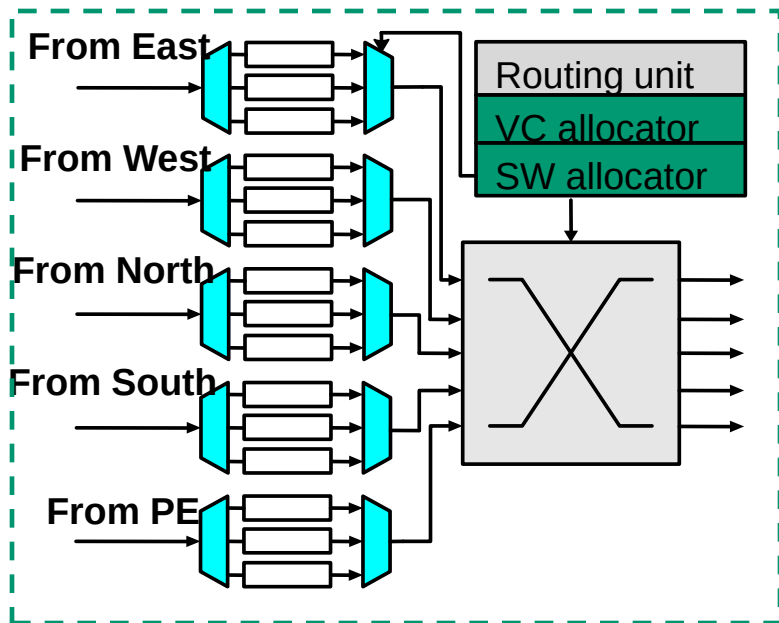


Wormhole Router Timeline

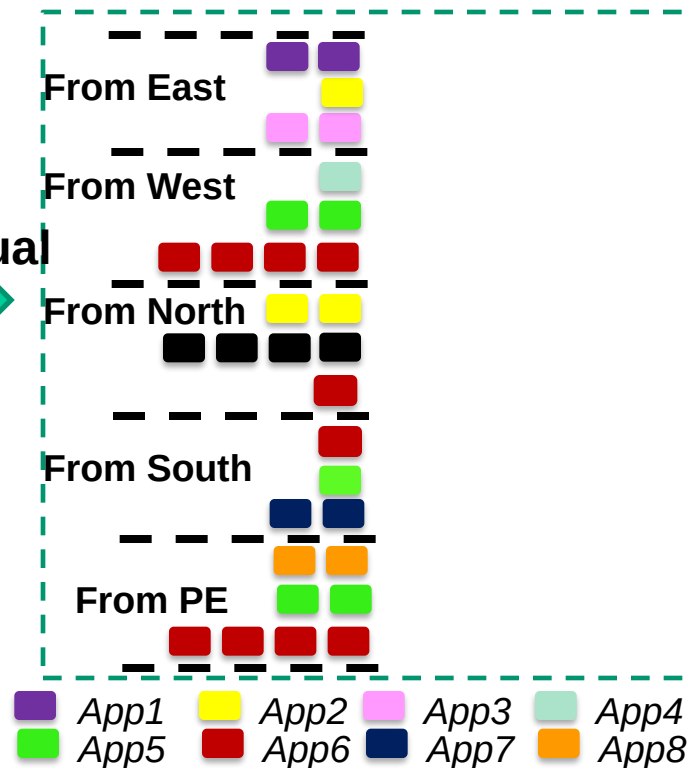


- ❖ Route computation performed once per packet
- ❖ Virtual channel allocated once per packet
- ❖ Body and tail flits inherit this information from head flit

Switch Level Packet Scheduling in NoC

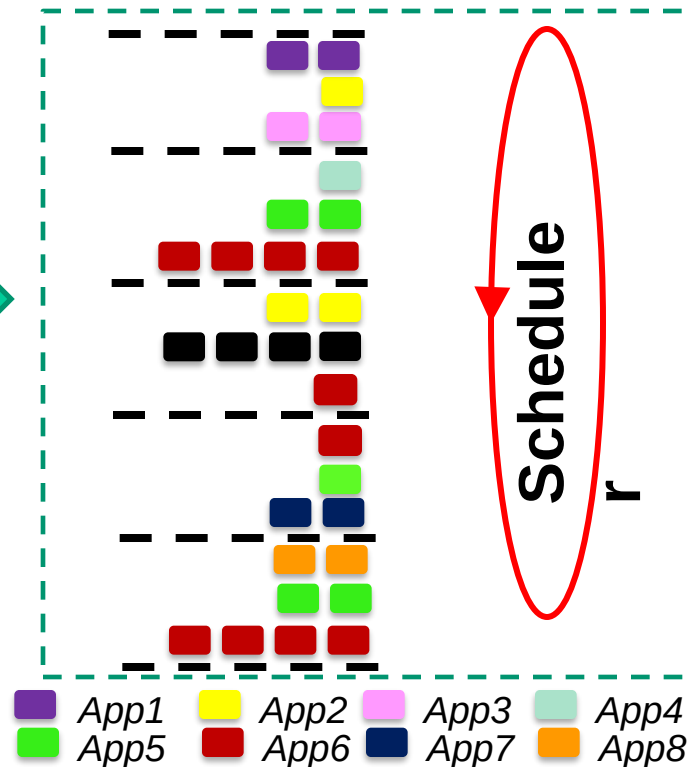
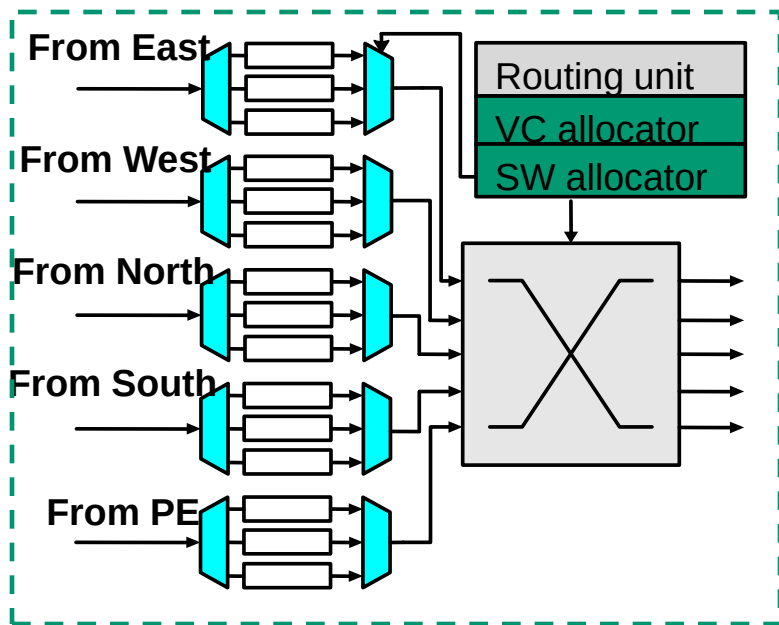


Conceptual
View

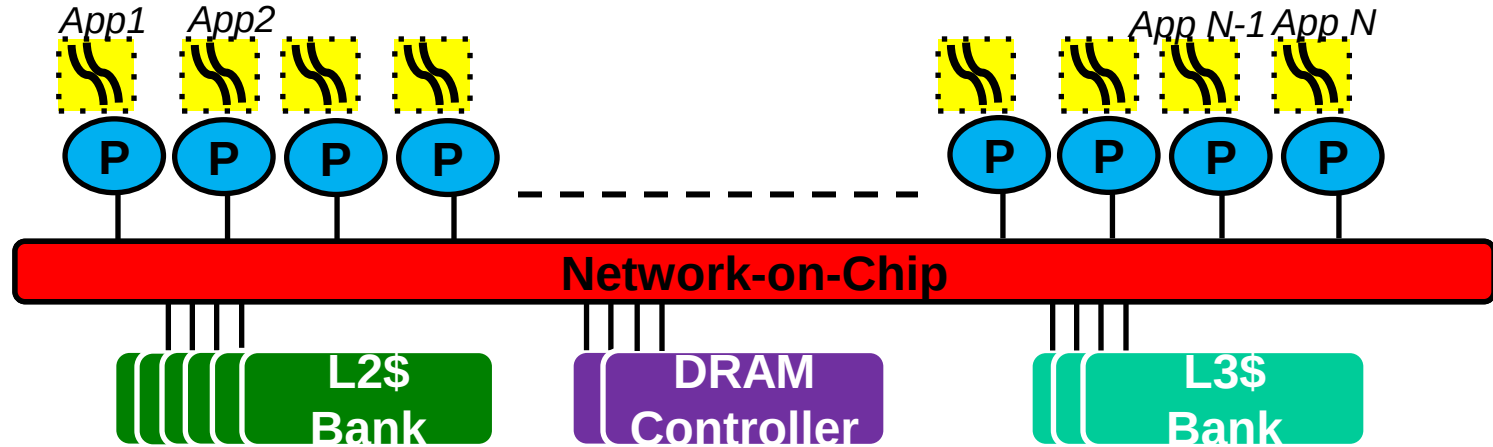


Input / Output Channel Selection

Which packet to choose?



How critical is NoC ?



Network-on-Chip is a **critical** resource
shared by **multiple applications**



johnjose@iitg.ac.in
<http://www.iitg.ac.in/johnjose/>